Discussion Section 4
STAT 132 , Winter 2025

1. Suppose that a random sample $X_1, \ldots, X_n$ is to be taken from the uniform distribution on the interval $[0, \theta]$ where $\theta > 0$ is unknown. How large must the sample size be to ensure that

$$\mathbb{P}(|\max\{X_1, \ldots, X_n\} - \theta| \leq 0.1\theta) \geq 0.95?$$

2. Suppose that a random sample is to be taken from the normal distribution with unknown mean μ and known variance $\sigma^2 = 4$. How large must the sample size be to ensure that

$$\mathbb{E}\left(|\bar{X}_n - \mu|^2\right) \leq 0.1?$$

3. Suppose that a random sample is to be taken from the normal distribution with unknown mean μ and known variance $\sigma^2 = 4$. How large must the sample size be to ensure that

$$\mathbb{P}(|\bar{X}_n - \mu| \leq 0.1) \geq 0.95?$$

4. Let $X_1, \ldots, X_n$ be a random sample from the exponential distribution with parameter λ. Find the CDF for the sampling distribution of the MLE of λ.

5. Suppose that $X_1, \ldots, X_n$ are a random sample from the Bernoulli distribution with unknown parameter p. How large must the sample size be to ensure that

$$\mathbb{E}(|\bar{X}_n - p|^2) \leq 0.01?$$